Retail Profitability and Discount Impact Analysis

Project Overview

This project audits 9,994 retail transactions ($2.30M in revenue) using MS Excel, MySQL, and Power BI to uncover why unprofitable orders are costing the business $156K+ in lost profit. Found out that discount over 20% destroy profit margin.

Dataset Summary

Rows: 9,994

Columns: 23

Key Features:

Order info: [row_id], [order_id], [order_date].

Customer details: [customer_id], [customer_name], [segment], [country], [city], [state], [postal_code], [region].

Product Details: [product_id], [category], [sub_category], [product_name].

Sales & Profit: [sales], [quantity], [discount], [profit], [cost_per_unit], [profit_margin], [discount_type].

Data Cleaning using Excel

- There were 0 null value in the dataset.
- The Shipping Date column has too many date inputs which are before the order date, so I dropped the column.
- Changed the order date column data from mat to “yyyy-mm-dd” to make it SQL ready.
- Created a new column [cost_per_unit] to calculate the sales price for product unit without discount.
- Created [profit_margin] column to calculate the % of profit per each order.
- Created [discount_type] column to group the % of discount into “No discount”(0%), “Low discount”(<20%), “Medium discount”(<40%) and “High discount”(>40%).
- Changed the column names to lower case and replaced the space (“ “) between the name with (_).
- Then uploaded this data set to MySQL using table input wizard.

EDA using MySQL

1. Total Sales:

total_sales
2297201.07

2. Total Profit:

total_profit
286397.62

3. Total Orders:

total_orders
9994

4. Average order value:

average_order_value
229.86

5. Total quantity sold:

total_quantity_sold
37873

6. Sales and Profit by region:

region	total_sales	total_profit
West	725457.93	108418.69
East	678781.36	91522.8
Central	501239.88	39706.44
South	391721.9	46749.69

7. Sales and Profit by category:

category	total_sales	total_profit
Technology	836154.1	145455.59
Furniture	741999.98	18451.25
Office Supplies	719046.99	122490.78

8. Top 10 Customers by Sales:

customer_name	total_sales
Sean Miller	25043.07
Tamara Chand	19052.22
Raymond Buch	15117.35
Tom Ashbrook	14595.62
Adrian Barton	14473.57
Ken Lonsdale	14175.23
Sanjit Chand	14142.34
Hunter Lopez	12873.3
Sanjit Engle	12209.44
Christopher Conant	12129.08

9. Top 10 Customers by Profit:

customer_name	total_profit
Tamara Chand	8981.32
Raymond Buch	6976.09
Sanjit Chand	5757.42
Hunter Lopez	5622.43
Adrian Barton	5444.81
Tom Ashbrook	4703.8
Christopher Martinez	3899.91
Keith Dawkins	3038.58
Andy Reiter	2884.61
Daniel Raglin	2869.08

10: Sales and Profit by month:

Year	Month	total_sales	total_profit
2016	April	28295.35	3488.86
2016	August	27909.47	5318.09
2016	December	69545.64	8983.57
2016	February	4519.92	862.3
2016	January	14236.9	2450.18
2016	July	33946.37	-841.46
2016	June	34595.14	4976.56
2016	March	55691.04	498.71
2016	May	23648.28	2738.74
2016	November	78628.74	9292.15
2016	October	31453.37	3448.22
2016	September	81777.34	8328.08
2017	April	34195.25	4187.53
2017	August	36898.32	5355.82
2017	December	74919.52	8016.95
2017	February	11951.4	2813.84
2017	January	18174.08	-3280.97
2017	July	28765.32	3288.71
2017	June	24797.31	3335.56
2017	March	38726.26	9732.07
2017	May	30131.72	4667.83
2017	November	75972.51	12474.79
2017	October	31404.9	2817.37
2017	September	64595.87	8209.16
2018	April	38750.04	2977.88
2018	August	31115.35	2062.04
2018	December	96999.07	17885.28
2018	February	22978.82	5004.58
2018	January	18542.52	2824.82
2018	July	39261.99	4432.9
2018	June	40344.54	4750.33
2018	March	51715.86	3611.97
2018	May	56987.75	8662.09
2018	November	79412.03	4011.45
2018	October	59687.8	16243.22
2018	September	73410.09	9328.67
2019	April	36521.52	933.3
2019	August	63120.85	9041.09
2019	December	83829.31	8483.38
2019	February	20301.12	1613.89
2019	January	43971.37	7140.44
2019	July	45264.43	6952.6
2019	June	52981.73	8223.41
2019	March	58872.35	14751.87
2019	May	44261.08	6342.7
2019	November	118447.81	9690.1
2019	October	77776.96	9275.38
2019	September	87866.66	10991.57

11. Top 5 product by sales

product_name	total_sales
Canon imageCLASS 2200 Advanced Copier	61599.83
Fellowes PB500 Electric Punch Plastic Comb Bind...	27453.38
Cisco TelePresence System EX90 Videoconferen...	22638.48
HON 5400 Series Task Chairs for Big and Tall	21870.57
GBC DocuBind TL300 Electric Binding System	19823.48

12. Top 5 product by profit

product_name	total_profit
Canon imageCLASS 2200 Advanced Copier	25199.94
Fellowes PB500 Electric Punch Plastic Comb Bind...	7753.06
Hewlett Packard LaserJet 3310 Copier	6983.89
Canon PC1060 Personal Laser Copier	4570.94
HP Designjet T520 Inkjet Large Format Printer -...	4094.98

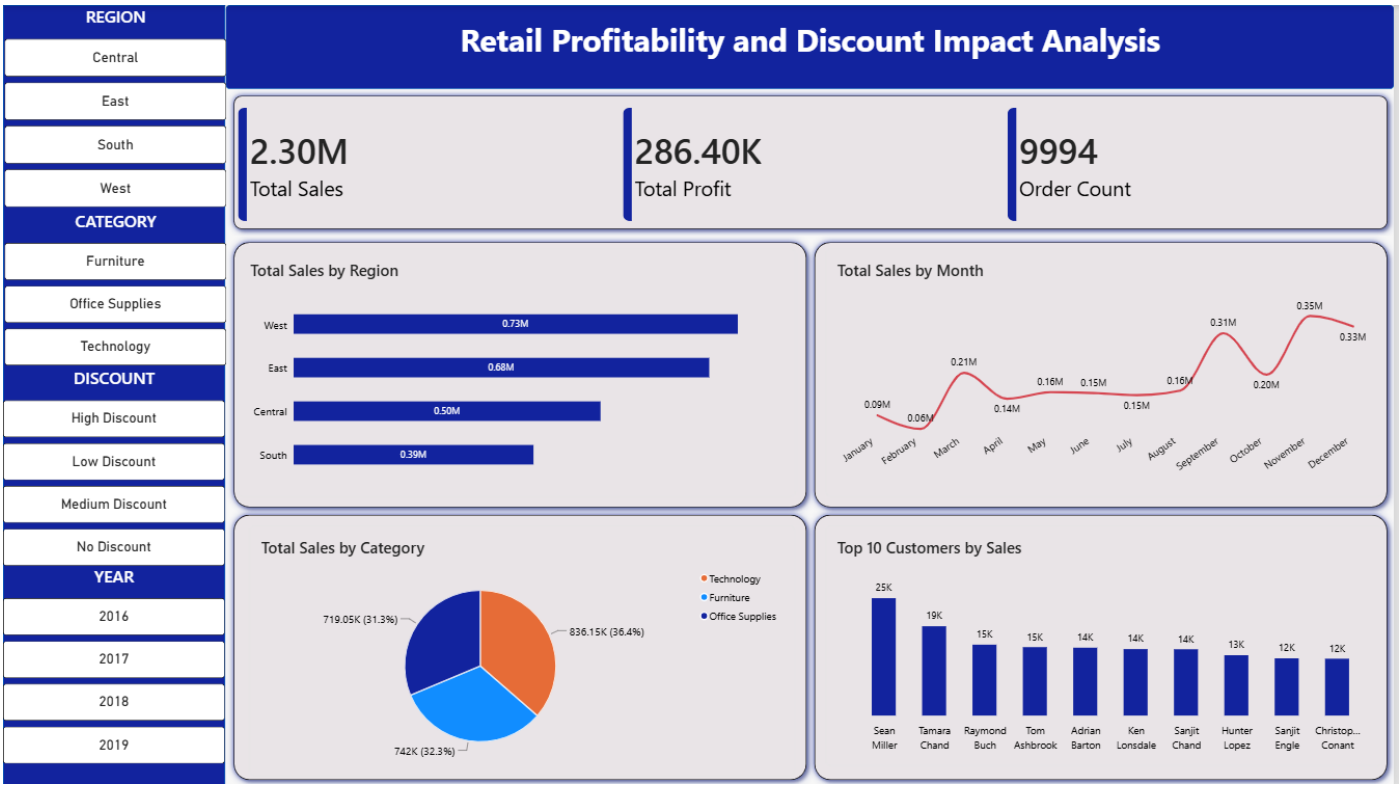
13. Discount Impact

discount_type	total_sales	total_profit
High Discount	128632.39	-99558.86
Medium Discount	234138.04	-35817.58
Low Discount	846522.17	100786.24
No Discount	1087908.47	320987.82

14. Number of orders by discount type

discount_type	total_orders
Medium Discount	460
High Discount	933
Low Discount	3803
No Discount	4798

Dashboard in Power BI



Business Insights

- The “West” region has the highest amount (725.46k) in sales and profit (108.42k).
- By category “Technology” generates highest in sales (836.15k) and profit (145.46k).
- Profits are highly influenced by discounts. While no discount (0%) and low discount (<20%) generates profits (421.77k), medium discount (<40%) and high discount (>40%) drives store to losses (-135.38k).
- Overall month of November performs best. But in different years such as in 2016 September, in 2017 November, in 2018 December and in 2019 November performed the best by sales.
- Furniture brings in huge sales but generates a tiny profit (2.49%) due to heavy discounts.
- Technology and Office Supplies carry the business. Technology made 17.4% profit margin and Office Supplies made 17% profit margin.

Business Recommendations

- Orders with 0% to 20% discounts make profits while discounts over 20% losses money almost every time. Cap all the stores discounts at 20%.
- Stop 70% to 80% clearance sales on binders and appliances. Deep clearance discounts caused almost 100k in direct losses. Replace 70% to 80% discount with “Buy 2 get 1 Free” bundles to clear old stock without losing money.
- Fix pricing on Tables and Bookcase, cause its loose money on almost every sale. Raise base prices by 10% or cut shipping subsidies on heavy furniture items.
- Texas, Ohio and Pennsylvania. These three states generate negative profit because of high discount. Reset their store pricing to match profitable states like California and New York.
- Reward sales persons based on profits, not just on sales.